Stored Energy in an Inductor: Solenoid Example

To make the idea of magnetic-field energy more concrete, let us consider a long solenoid. This gives a simple circuit-level interpretation of the field-energy result derived earlier.

Long solenoid and its inductance

Consider a solenoid of length ℓ, cross-sectional area A, and total number of turns N. If the turns per unit length are denoted by

$$n = \frac{N}{\ell}, \tag{1}$$

then, for a long solenoid, the magnetic field inside is approximately

$$\vec{B} \approx \mu n I \, \hat{z}. \tag{2}$$

Hence the magnetic flux through each turn is

$$\Phi_B = BA = \mu n I A. \tag{3}$$

The total flux linkage is

$$N\Phi_B = N(\mu n I A). \tag{4}$$

By definition of inductance,

$$N\Phi_B = LI. \tag{5}$$

Therefore,

$$L = \frac{N\Phi_B}{I} = N\mu n A. \tag{6}$$

Using $N = n\ell$, we obtain

$$\boxed{L = \mu n^2 A \ell} \tag{7}$$

Why must the source do work?

When the current in the solenoid is increased, the magnetic field inside the solenoid also increases. Hence the magnetic flux through each turn increases with time.

By Faraday's law, an induced emf appears which opposes that increase:

$$\mathcal{E}_{\text{ind}} = -N\frac{d\Phi_B}{dt}. \tag{8}$$

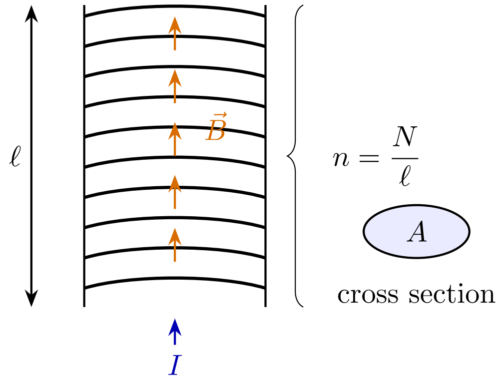

Figure 1: A long solenoid of length ℓ, cross-sectional area A, and turns per unit length n.

Since $N\Phi_B = LI$,

$$\mathcal{E}_{\text{ind}} = -L\frac{dI}{dt}. \tag{9}$$

Thus, to increase the current, the external source must overcome this induced emf. Under the usual passive sign convention, the voltage across an ideal inductor is written as

$$\boxed{v_L = L\frac{dI}{dt}} \tag{10}$$

This expression tells us that a changing current requires an applied voltage.

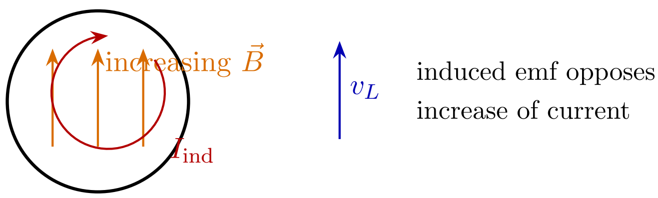

Figure 2: As the magnetic field increases, the induced emf opposes the increase of current. Hence an external voltage is required to build up the current.

Energy supplied to the inductor

The instantaneous power delivered to the inductor is

$$p = v_L I. \tag{11}$$

Using $v_L = L\frac{dI}{dt}$,

$$p = LI\frac{dI}{dt}. \tag{12}$$

Therefore, the energy supplied from time 0 to time t is

$$U_B(t) = \int_0^t p\,dt = \int_0^t LI\frac{dI}{dt}\,dt. \tag{13}$$

Assuming L is constant,

$$U_B(t) = L\int_0^{I(t)} I\,dI. \tag{14}$$

Hence,

$$\boxed{U_B(t) = \frac{1}{2}LI^2(t)} \tag{15}$$

More generally, if the current changes from I_1 to I_2, then

$$\boxed{\Delta U_B = \frac{L}{2}\left(I_2^2 - I_1^2\right)} \tag{16}$$

So, when the current is built up from 0 to I,

$$\boxed{U_B = \frac{1}{2}LI^2} \tag{17}$$

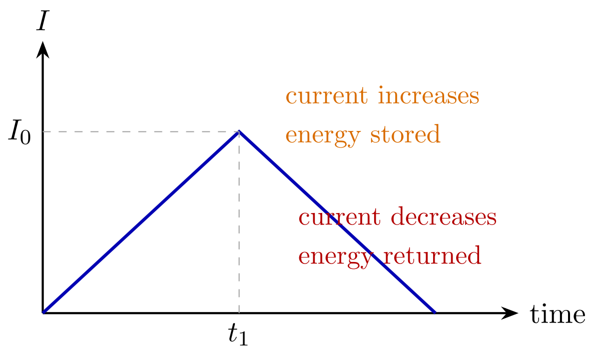

Figure 3: In an ideal inductor, energy is stored when the current increases and returned when the current decreases.

Why an inductor is not dissipative

The result

$$\Delta U_B = \frac{L}{2}\left(I_2^2 - I_1^2\right) \tag{18}$$

shows that the energy depends only on the initial and final currents.

For example, if the current is increased from 0 to I_0 and then brought back again from I_0 to 0, the net change in stored energy is

$$\Delta U_B = \frac{L}{2}(I_0^2 - 0) + \frac{L}{2}(0 - I_0^2) = 0. \tag{19}$$

So, in an ideal inductor, the energy is not dissipated; it is merely stored and later returned. Thus an ideal inductor is a **non-dissipative** element.

Important Interpretation

An ideal resistor dissipates energy as heat, but an ideal inductor does not. The energy supplied to an ideal inductor during current buildup is stored in its magnetic field and can be completely recovered later.

Connection with magnetic field energy density

Now let us connect the circuit result to the field result.

For the solenoid,

$$U_B = \frac{1}{2}LI^2 = \frac{1}{2}(\mu n^2 A\ell)I^2. \tag{20}$$

But inside the solenoid,

$$B = \mu n I. \tag{21}$$

Therefore,

$$U_B = \frac{1}{2}\mu n^2 I^2 A\ell = \frac{1}{2}\frac{B^2}{\mu}A\ell. \tag{22}$$

Since $A\ell$ is the volume of the magnetic-field region inside the solenoid,

$$\boxed{U_B = \frac{B^2}{2\mu} \times \text{volume}} \tag{23}$$

Hence the magnetic energy density is

$$\boxed{u_B = \frac{B^2}{2\mu}} \tag{24}$$

This agrees exactly with the general field-energy expression derived earlier:

$$U_B = \iiint_V \frac{B^2}{2\mu}\,dv \qquad \text{(for a linear magnetic medium)}. \tag{25}$$

Boxed Summary

For a long solenoid:

1. $$B = \mu n I, \qquad \Phi_B = BA = \mu n I A$$

2. Inductance:
$$L = \frac{N\Phi_B}{I} = \mu n^2 A \ell$$

3. Induced emf:
$$\mathcal{E}_{\text{ind}} = -L\frac{dI}{dt}$$
and the applied inductor voltage is
$$v_L = L\frac{dI}{dt}$$

4. Power delivered to the inductor:
$$p = v_L I = LI\frac{dI}{dt}$$

5. Energy stored:
$$U_B = \frac{1}{2}LI^2$$

6. Since $B = \mu n I$,
$$U_B = \frac{B^2}{2\mu} \times \text{volume}$$
so the magnetic energy density is
$$\boxed{u_B = \frac{B^2}{2\mu}}$$

Thus, the circuit-level formula $\frac{1}{2}LI^2$ and the field-level formula $\iiint \frac{B^2}{2\mu}\,dv$ are fully consistent.

Ampère's Law with Maxwell's Correction

So far, in magnetostatics, we used Ampère's law in the form

$$\nabla \times \vec{H} = \vec{J}_{\text{free}}. \tag{26}$$

This works well for *steady currents*. However, when fields vary with time, Eq. (26) is incomplete. We now understand why a correction is needed, and how Maxwell arrived at the modified form.

A caution: do not over-interpret an isolated current element

One may be tempted to start from the Biot–Savart law for a tiny current element $I\,d\vec{\ell}$ at the origin:

$$d\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi}\frac{I\,d\vec{\ell} \times \hat{r}}{r^2}. \tag{27}$$

For example, if $d\vec{\ell} = d\ell\,\hat{z}$, then

$$d\vec{B}(\vec{r}) = \frac{\mu_0 I d\ell}{4\pi}\frac{x\,\hat{y} - y\,\hat{x}}{r^3}, \qquad r = \sqrt{x^2+y^2+z^2}. \tag{28}$$

If one directly computes the curl of this expression, one may get misleading intermediate expressions suggesting $\nabla \times d\vec{B} \neq 0$ away from the origin. But there is an important conceptual issue here:

Important caution

An isolated current element is **not a physically realizable steady-current distribution**. Current must flow in a closed path (or satisfy charge continuity). Therefore, Ampère's law in magnetostatics should be applied to **complete steady current distributions**, not to an isolated open current segment.

So the apparent contradiction is not that Biot–Savart's law is wrong or that Ampère's law is wrong; rather, the "isolated current element" is not a valid complete magnetostatic source by itself.

Steady closed currents: a small current loop

A physically meaningful localized source is a *small current loop*. Far away from the loop, its magnetic field has the same angular form as the electric field of an electric dipole.

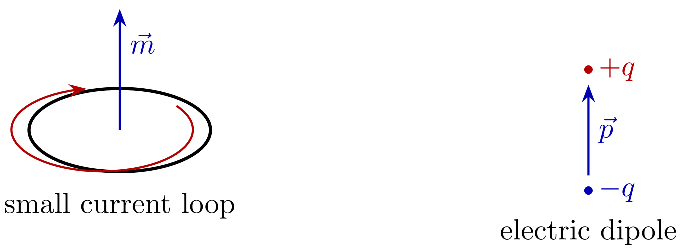

Figure 4: A small current loop behaves like a magnetic dipole with moment $\vec{m}$, analogous in form to an electric dipole with moment $\vec{p}$.

For a small current loop, the magnetic field in the source-free region has dipole form,

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi r^3}\left(2m\cos\theta\,\hat{r} + m\sin\theta\,\hat{\theta}\right), \tag{29}$$

which is analogous to the electric dipole field

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0 r^3}\left(2p\cos\theta\,\hat{r} + p\sin\theta\,\hat{\theta}\right). \tag{30}$$

Since in electrostatics we know that $\vec{E} = -\nabla\phi$, it follows that

$$\nabla \times \vec{E} = \nabla \times (-\nabla\phi) = 0 \tag{31}$$

everywhere away from the charges.

Likewise, for a *steady closed current distribution*, we expect

$$\nabla \times \vec{H} = 0 \tag{32}$$

at points where no current is present. So magnetostatics is consistent when applied properly to *steady closed currents*.

Side Note: Why a current element behaves like a time-varying electric dipole

An isolated current element $I\,d\vec{\ell}$ is not a physically valid *steady* current by itself, because current continuity would be violated at its ends. To understand what such an element really represents, imagine a very short wire segment of length $d\ell$ carrying current I upward for a short time dt.
In time dt, the amount of charge transported is

$$dq = I\,dt. \tag{33}$$

This means that one end of the small segment loses charge dq, while the other end gains charge dq.
So the current element effectively creates a small charge separation, i.e. an electric dipole.
If the separation vector is $d\vec{\ell}$, then the dipole moment formed in time dt is

$$d\vec{p} = dq\,d\vec{\ell} = I\,dt\,d\vec{\ell}. \tag{34}$$

Therefore,

$$\boxed{\frac{d\vec{p}}{dt} = I\,d\vec{\ell}} \tag{35}$$

So a current element $I\,d\vec{\ell}$ may be interpreted as the **time rate of change of an electric dipole moment**.

Meaning:

- A *closed steady current loop* is a proper magnetostatic source.
- An *isolated current element* behaves instead like a **time-varying electric dipole source**.
- That is why using the magnetostatic form of Ampère's law blindly for an isolated current element can be misleading.

Where the problem really appears: time-varying charge

The true limitation of Eq. (26) appears when the current is not steady.

If we take the divergence of Eq. (26),

$$\nabla \cdot (\nabla \times \vec{H}) = \nabla \cdot \vec{J}_{\text{free}}. \tag{36}$$

But the divergence of a curl is always zero:

$$\nabla \cdot (\nabla \times \vec{H}) = 0. \tag{37}$$

Hence Eq. (26) implies

$$\nabla \cdot \vec{J}_{\text{free}} = 0. \tag{38}$$

However, from the continuity equation,

$$\nabla \cdot \vec{J}_{\text{free}} = -\frac{\partial \rho_{\text{free}}}{\partial t}. \tag{39}$$

So Eq. (38) would force

$$\frac{\partial \rho_{\text{free}}}{\partial t} = 0. \tag{40}$$

This means the original Ampère law is valid only when the charge density is not changing with time, i.e. for **steady-state magnetostatics**. It cannot be the complete law for general time-varying situations.

Charging capacitor: the key physical example

Consider a capacitor being charged by a current I.

Take the same closed path C, but choose two different surfaces bounded by it:

- S_1: cuts the wire, so the enclosed conduction current is I.
- S_2: passes through the capacitor gap, so no conduction current crosses it.

If we use the old Ampère law in integral form,

$$\oint_C \vec{H} \cdot d\vec{l} = I_{\text{free,enclosed}}, \tag{41}$$

then the same left-hand side would give two different answers:

$$I \quad \text{using } S_1, \qquad 0 \quad \text{using } S_2.$$

This is impossible.

So something is missing in the capacitor gap.

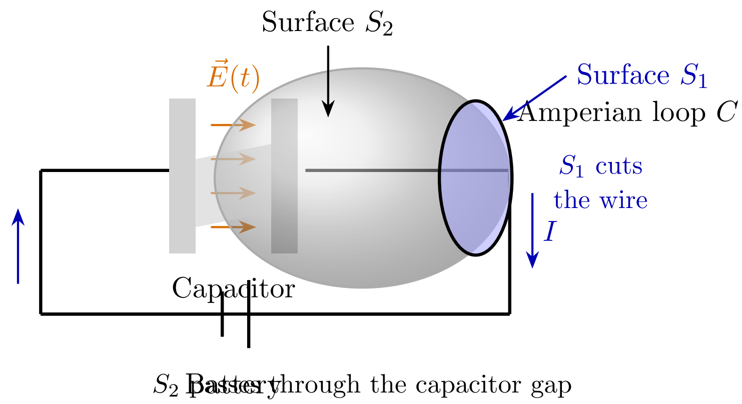

Figure 5: A charging capacitor illustrating two different surfaces with the same boundary loop C. The flat surface S_1 cuts the wire and encloses conduction current, while the bulged surface S_2 passes through the capacitor gap, where the changing electric field gives rise to displacement current.

Maxwell's idea: add displacement current

Between the capacitor plates, although no conduction current flows, the electric field changes with time. This changing electric field must contribute to the magnetic effect.

Recall Gauss's law in matter:

$$\nabla \cdot \vec{D} = \rho_{\text{free}}. \tag{42}$$

Differentiate with respect to time:

$$\nabla \cdot \frac{\partial \vec{D}}{\partial t} = \frac{\partial \rho_{\text{free}}}{\partial t}. \tag{43}$$

Using the continuity equation,

$$\nabla \cdot \vec{J}_{\text{free}} = -\frac{\partial \rho_{\text{free}}}{\partial t}, \tag{44}$$

we get

$$\nabla \cdot \left(\vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t} \right) = 0. \tag{45}$$

This suggests the corrected law

$$\boxed{\nabla \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}} \tag{46}$$

The extra term

$$\boxed{\vec{J}_d = \frac{\partial \vec{D}}{\partial t}} \tag{47}$$

is called the **displacement current density**.

> **Key idea of Maxwell's correction**
>
> A time-varying electric field behaves, in Ampère's law, as if it were an additional current source. This is **not** conduction current; it is called **displacement current**.

Integral form of the modified law

Integrating Eq. (46) over a surface S bounded by C, and applying Stokes' theorem:

$$\oint_C \vec{H} \cdot d\vec{l} = \iint_S \left(\vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t} \right) \cdot d\vec{a}. \tag{48}$$

Hence,

$$\boxed{\oint_C \vec{H} \cdot d\vec{l} = I_{\text{free,enclosed}} + \frac{d}{dt} \iint_S \vec{D} \cdot d\vec{a}} \tag{49}$$

In free space, where $\vec{D} = \epsilon_0 \vec{E}$, this becomes

$$\boxed{\oint_C \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} + \mu_0 \epsilon_0 \frac{d}{dt} \iint_S \vec{E} \cdot d\vec{a}} \tag{50}$$

or equivalently,

$$\boxed{\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}} \tag{51}$$

Displacement current in a charging capacitor

For the capacitor gap,

$$I_d = \frac{d}{dt} \iint_S \vec{D} \cdot d\vec{a}. \tag{52}$$

In a simple parallel-plate capacitor, the electric flux changes because charge accumulates on the plates. Using

$$\iint_S \vec{D} \cdot d\vec{a} = Q_{\text{free}}, \tag{53}$$

we get

$$I_d = \frac{dQ_{\text{free}}}{dt} = I. \tag{54}$$

So the displacement current between the plates is exactly equal to the conduction current in the wires. That is why the result is now independent of which surface we choose.

Charging capacitor result

For a charging capacitor,

$$I_d = \frac{d}{dt} \iint_S \vec{D} \cdot d\vec{a} = I.$$

Hence Ampère–Maxwell law gives the same answer for every surface bounded by the same contour.

Physical meaning

The original Ampère law was incomplete because it only accounted for conduction current. Maxwell realized that a changing electric field must also produce magnetic effects. This correction is crucial because:

1. it restores compatibility with the continuity equation,
2. it resolves the charging-capacitor paradox,
3. it makes electromagnetism self-consistent,
4. and it ultimately predicts electromagnetic waves.

Comparison: Ampère law before and after Maxwell

Magnetostatic form:

$$\nabla \times \vec{H} = \vec{J}_{\text{free}} \qquad \text{(valid for steady currents)}$$

General time-varying form:

$$\nabla \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}$$

The extra term

$$\frac{\partial \vec{D}}{\partial t}$$

is the **displacement current density**.

Boxed Summary

1. The original Ampère law
$$\nabla \times \vec{H} = \vec{J}_{\text{free}}$$
implies $\nabla \cdot \vec{J}_{\text{free}} = 0$, so it is valid only for steady currents.

2. In a charging capacitor, the same contour can bound two different surfaces: one encloses conduction current, the other does not. This shows that the old law is incomplete.

3. Maxwell corrected Ampère's law by adding the displacement-current term:
$$\nabla \times \vec{H} = \vec{J}_{\text{free}} + \frac{\partial \vec{D}}{\partial t}.$$

4. The integral form is
$$\oint_C \vec{H} \cdot d\vec{l} = I_{\text{free,enclosed}} + \frac{d}{dt} \iint_S \vec{D} \cdot d\vec{a}.$$

5. In free space:
$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}.$$

6. For a charging capacitor,
$$I_d = \frac{d}{dt} \iint_S \vec{D} \cdot d\vec{a} = I,$$
so the law becomes surface-independent and self-consistent.

Why Maxwell's Correction Was So Important

We may now collect the four fundamental equations of classical electromagnetics:

$$\nabla \cdot \vec{D} = \rho_v, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}. \tag{55}$$

Maxwell's Equations

$$\nabla \cdot \vec{D} = \rho_v \qquad \text{(Gauss's law for electricity)}$$

$$\nabla \cdot \vec{B} = 0 \qquad \text{(Gauss's law for magnetism)}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \qquad \text{(Faraday's law)}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \qquad \text{(Ampère–Maxwell law)}$$

The last equation tells us something remarkable:

> *Not only can a changing magnetic field induce an electric field, but a changing electric field can also induce a magnetic field.*

This is the key step that turns separate electrostatics and magnetostatics into a unified theory of **electromagnetism**.

Connection with energy flow

Recall the expression that appeared while deriving magnetic-field energy:

$$U_B = \int_0^T \iiint_V \left[\nabla \cdot (\vec{E} \times \vec{H}) - \vec{H} \cdot \frac{\partial \vec{B}}{\partial t} \right] dv\, dt. \tag{56}$$

In the quasistatic magnetic-energy discussion, we neglected the surface term

$$\iiint_V \nabla \cdot (\vec{E} \times \vec{H})\, dv = \oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a},$$

because for localized non-radiating situations it vanishes at a very large enclosing surface.

However, after Maxwell's correction, this term becomes deeply important in general time-varying problems. It represents **energy flow carried by the electromagnetic field**.

$$\boxed{\vec{S} = \vec{E} \times \vec{H}} \tag{57}$$

is called the **Poynting vector**.

Meaning of the Poynting Vector

$$\vec{S} = \vec{E} \times \vec{H}$$

gives the **electromagnetic power flow per unit area**. Its SI unit is

$$\frac{\mathrm{W}}{\mathrm{m}^2}.$$

The direction of $\vec{S}$ gives the direction in which electromagnetic energy is transported.

Why this surface term was negligible earlier

In electrostatics, the electric field of a localized charge distribution falls as

$$E \sim \frac{1}{r^2},$$

while in magnetostatics the magnetic field of a localized current distribution also falls roughly as

$$H \sim \frac{1}{r^2}.$$

Hence, for such non-radiating static fields,

$$|\vec{E} \times \vec{H}| \sim \frac{1}{r^4}. \tag{58}$$

Therefore, over a spherical surface of area proportional to r^2,

$$\oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \sim \frac{1}{r^2} \to 0 \qquad (r \to \infty). \tag{59}$$

That is why this term was dropped in the earlier derivation of magnetic energy storage.

But in electromagnetic radiation...

In a radiating electromagnetic wave,

$$E \sim \frac{1}{r}, \qquad H \sim \frac{1}{r}.$$

Hence,

$$|\vec{E} \times \vec{H}| \sim \frac{1}{r^2}. \tag{60}$$

Now, when we integrate over a sphere of area proportional to r^2, we get a *finite nonzero result*:

$$\oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \neq 0. \tag{61}$$

This means energy is genuinely leaving the source and propagating outward through space.

Important Distinction

- For localized static or quasistatic fields:

  $$\oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \to 0$$

  at infinity.

- For radiating electromagnetic waves:

  $$\oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \neq 0.$$

So the surface term that was negligible before now becomes the mathematical signature of **electromagnetic radiation**.

Radiation from a distant source

A star is a familiar example of a source radiating electromagnetic energy. The emitted energy spreads over larger and larger spherical surfaces as it propagates outward.

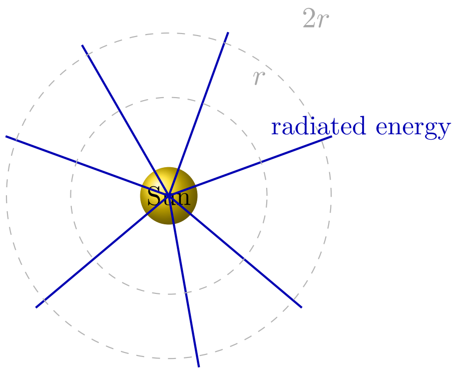

Figure 6: Radiation from a star. Electromagnetic energy propagates outward and spreads over larger spherical surfaces.

If the same total power is spread over a sphere of radius r, then the area is

$$4\pi r^2.$$

Hence the power per unit area decreases as

$$\boxed{|\vec{S}| \propto \frac{1}{r^2}.} \tag{62}$$

This is the familiar inverse-square law for intensity.

Why Maxwell's correction unified electricity, magnetism, and optics

Before Maxwell, electricity and magnetism were closely related, but optics was treated as a separate subject. Maxwell's correction completed the symmetry:

- a changing magnetic field creates an electric field,
- a changing electric field creates a magnetic field.

This mutual coupling allows self-sustaining waves:

$$\vec{E} \longleftrightarrow \vec{B}.$$

Such coupled disturbances can propagate through space even in the absence of charges and currents. These are **electromagnetic waves**.

Therefore, visible light, radio waves, microwaves, infrared, ultraviolet, X-rays, and gamma rays are all just different forms of electromagnetic waves.

Historical Importance

Maxwell's displacement-current term did much more than fix Ampère's law. It led to the prediction that light itself is an electromagnetic wave. Thus, **optics became part of electromagnetics**.

Energy transport by the electromagnetic field

The quantity

$$\vec{S} = \vec{E} \times \vec{H}$$

is called the Poynting vector. It tells us how much electromagnetic energy crosses a unit area per unit time.

Thus:

$$\boxed{\text{Power crossing surface } S = \iint_S (\vec{E} \times \vec{H}) \cdot d\vec{a}} \tag{63}$$

So when

$$\iint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \neq 0,$$

the field is carrying energy across that surface.

Boxed Summary

1. Maxwell's equations are

$$\nabla \cdot \vec{D} = \rho_v, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}.$$

2. Maxwell's correction means that a **changing electric field also produces magnetic effects**.

3. The quantity

$$\vec{S} = \vec{E} \times \vec{H}$$

is the **Poynting vector**, representing electromagnetic power flow per unit area.

4. For static localized fields,

$$E \sim \frac{1}{r^2}, \quad H \sim \frac{1}{r^2} \quad \Rightarrow \quad \oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \to 0.$$

5. For electromagnetic radiation,

$$E \sim \frac{1}{r}, \quad H \sim \frac{1}{r} \quad \Rightarrow \quad \oiint_S (\vec{E} \times \vec{H}) \cdot d\vec{a} \neq 0.$$

6. Thus Maxwell's correction not only makes Ampère's law self-consistent, but also explains **electromagnetic waves** and unifies **electricity, magnetism, and optics**.

Introduction to the Wave Equation

We now show how Maxwell's equations themselves predict the existence of waves in free space.

Assume a simple field configuration

Let us assume that the electric and magnetic fields have the form

$$\vec{E} = E_0\,\hat{x}\,f(z,t), \qquad \vec{B} = B_0\,\hat{y}\,g(z,t), \tag{64}$$

where $f(z,t)$ and $g(z,t)$ are functions to be determined.

Thus:

- the electric field is along $\hat{x}$,
- the magnetic field is along $\hat{y}$,
- both vary only with z and t.

We take the medium to be **free space**, so that

$$\rho_v = 0, \qquad \vec{J} = 0. \tag{65}$$

Hence Maxwell's equations reduce to

$$\nabla \cdot \vec{E} = 0, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}. \tag{66}$$

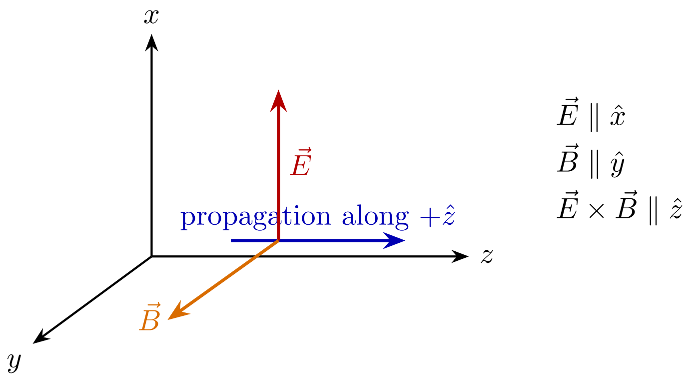

Figure 7: A plane electromagnetic wave in free space: $\vec{E}$, $\vec{B}$, and the direction of propagation are mutually perpendicular.

Divergence equations

From Eq. (64),

$$\vec{E} = E_0 f(z,t)\,\hat{x}, \qquad \vec{B} = B_0 g(z,t)\,\hat{y}. \tag{67}$$

Therefore,

$$\nabla \cdot \vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 0, \tag{68}$$

because E_x depends only on z, t and $E_y = E_z = 0$.

Similarly,

$$\nabla \cdot \vec{B} = 0. \tag{69}$$

So the assumed fields automatically satisfy the divergence equations.

Faraday's law

Now evaluate

$$\nabla \times \vec{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ E_0 f(z,t) & 0 & 0 \end{vmatrix}. \tag{70}$$

Hence,

$$\nabla \times \vec{E} = E_0 \frac{\partial f(z,t)}{\partial z}\,\hat{y}. \tag{71}$$

Also,

$$-\frac{\partial \vec{B}}{\partial t} = -B_0 \frac{\partial g(z,t)}{\partial t}\,\hat{y}. \tag{72}$$

Therefore, from Faraday's law,

$$E_0 \frac{\partial f}{\partial z} = -B_0 \frac{\partial g}{\partial t}. \tag{73}$$

Ampère–Maxwell law

Now evaluate

$$\nabla \times \vec{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 0 & B_0 g(z,t) & 0 \end{vmatrix}. \tag{74}$$

So,

$$\nabla \times \vec{B} = -B_0 \frac{\partial g(z,t)}{\partial z}\,\hat{x}. \tag{75}$$

Also,

$$\mu_0\epsilon_0 \frac{\partial \vec{E}}{\partial t} = \mu_0\epsilon_0 E_0 \frac{\partial f(z,t)}{\partial t}\,\hat{x}. \tag{76}$$

Thus Ampère–Maxwell law gives

$$-B_0 \frac{\partial g}{\partial z} = \mu_0\epsilon_0 E_0 \frac{\partial f}{\partial t}. \tag{77}$$

Deriving the wave equation for the electric field

Differentiate Eq. (73) with respect to z:

$$E_0 \frac{\partial^2 f}{\partial z^2} = -B_0 \frac{\partial}{\partial z}\left(\frac{\partial g}{\partial t}\right). \tag{78}$$

Since mixed partial derivatives commute,

$$E_0 \frac{\partial^2 f}{\partial z^2} = -B_0 \frac{\partial}{\partial t}\left(\frac{\partial g}{\partial z}\right). \tag{79}$$

Now from Eq. (77),

$$\frac{\partial g}{\partial z} = -\frac{\mu_0\epsilon_0 E_0}{B_0}\frac{\partial f}{\partial t}. \tag{80}$$

Substituting,

$$E_0 \frac{\partial^2 f}{\partial z^2} = -B_0 \frac{\partial}{\partial t}\left(-\frac{\mu_0\epsilon_0 E_0}{B_0}\frac{\partial f}{\partial t}\right). \tag{81}$$

Therefore,

$$E_0 \frac{\partial^2 f}{\partial z^2} = \mu_0\epsilon_0 E_0 \frac{\partial^2 f}{\partial t^2}. \tag{82}$$

Cancelling E_0,

$$\boxed{\frac{\partial^2 f(z,t)}{\partial z^2} = \mu_0\epsilon_0 \frac{\partial^2 f(z,t)}{\partial t^2}} \tag{83}$$

This is the **wave equation** for the electric field amplitude function.

Hence,

$$\boxed{\frac{\partial^2 E_x(z,t)}{\partial z^2} = \mu_0\epsilon_0 \frac{\partial^2 E_x(z,t)}{\partial t^2}} \tag{84}$$

Wave equation for the magnetic field

In the same way, starting from Eq. (73) and Eq. (77), one can show that

$$\boxed{\frac{\partial^2 g(z,t)}{\partial z^2} = \mu_0\epsilon_0 \frac{\partial^2 g(z,t)}{\partial t^2}} \tag{85}$$

or equivalently,

$$\boxed{\frac{\partial^2 B_y(z,t)}{\partial z^2} = \mu_0\epsilon_0 \frac{\partial^2 B_y(z,t)}{\partial t^2}} \tag{86}$$

So both $\vec{E}$ and $\vec{B}$ satisfy the same type of wave equation.

Wave speed

The standard one-dimensional wave equation is

$$\frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2}\frac{\partial^2 \psi}{\partial t^2}. \tag{87}$$

Comparing with Eq. (83) or Eq. (85),

$$\frac{1}{v^2} = \mu_0\epsilon_0. \tag{88}$$

Hence,

$$\boxed{v = \frac{1}{\sqrt{\mu_0\epsilon_0}}} \tag{89}$$

But this value is exactly the speed of light in free space:

$$\boxed{c = \frac{1}{\sqrt{\mu_0\epsilon_0}}} \tag{90}$$

Thus Maxwell concluded that **light is an electromagnetic wave**.

Historic Conclusion

Maxwell's equations predict waves that travel in free space with speed

$$c = \frac{1}{\sqrt{\mu_0\epsilon_0}}.$$

Since this equals the measured speed of light, Maxwell identified light as an electromagnetic wave.

General form of the solution

The general solution of the one-dimensional wave equation is

$$f(z,t) = F(z-ct) + G(z+ct), \tag{91}$$

where

- $F(z-ct)$ represents a wave travelling in the $+\hat{z}$ direction,
- $G(z+ct)$ represents a wave travelling in the $-\hat{z}$ direction.

Thus,

$$E_x(z,t) = F(z-ct) + G(z+ct), \tag{92}$$

and similarly for $B_y(z,t)$.

Important clarification

The solution obtained here corresponds to an **ideal plane wave**. Its amplitude does not decrease with z because the wavefront is assumed infinite and planar.

This does *not* contradict the inverse-square law for radiation from localized sources. That law applies to **spherical spreading** from a point-like source, whereas here we are dealing with a **plane-wave model**.

Plane wave vs. spherical wave

- **Plane wave:** idealized wave; amplitude does not decay with z.
- **Spherical wave:** emitted by a localized source; amplitude falls approximately as $1/r$.

So the present derivation gives the **local plane-wave form** of electromagnetic propagation in free space.

Boxed Summary

Assuming

$$\vec{E} = E_0 \hat{x}\, f(z,t), \qquad \vec{B} = B_0 \hat{y}\, g(z,t)$$

in free space, Maxwell's equations give

$$E_0 \frac{\partial f}{\partial z} = -B_0 \frac{\partial g}{\partial t}, \qquad -B_0 \frac{\partial g}{\partial z} = \mu_0 \epsilon_0 E_0 \frac{\partial f}{\partial t}.$$

Eliminating g (or f) leads to the wave equations

$$\frac{\partial^2 E_x}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 E_x}{\partial t^2}, \qquad \frac{\partial^2 B_y}{\partial z^2} = \mu_0 \epsilon_0 \frac{\partial^2 B_y}{\partial t^2}.$$

Therefore electromagnetic waves propagate in free space with speed

$$\boxed{c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}}$$

and hence light is an electromagnetic wave.

Concluding Remarks

We may now summarize the central outcome of the course.

Maxwell's Equations

In free space, Maxwell's equations are

$$\nabla \cdot \vec{E} = \frac{\rho_v}{\epsilon_0}, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}. \tag{93}$$

In material media, they are more commonly written as

$$\nabla \cdot \vec{D} = \rho_f, \qquad \nabla \cdot \vec{B} = 0, \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \qquad \nabla \times \vec{H} = \vec{J}_f + \frac{\partial \vec{D}}{\partial t}. \tag{94}$$

Lorentz Force Law

The effect of the electromagnetic field on a charge is described by the Lorentz force law:

$$\boxed{\vec{F} = q\vec{E} + q(\vec{v} \times \vec{B})} \tag{95}$$

Course Summary: What the equations mean

- **Maxwell's equations** tell us how **charges and currents produce electric and magnetic fields**.
- **Lorentz force law** tells us how **electric and magnetic fields act on charges**.

Big Picture

Together, **Maxwell's equations** and the **Lorentz force law** summarize the entire framework of **classical electromagnetics**.

Final Takeaway from EE1204

Throughout this course, we have seen a beautiful progression:

1. charges produce electric fields,
2. currents produce magnetic fields,
3. changing magnetic fields induce electric fields,
4. changing electric fields induce magnetic fields,
5. and these coupled fields propagate as **electromagnetic waves**.

Thus, electricity, magnetism, and light are not separate subjects, but different manifestations of one unified theory: **electromagnetics**.